

Rethinking Acoustic Sensing: Challenges and Perspectives for Wearable Sensing in the wild

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Abstract

The ubiquitous presence of acoustic-enabled smart devices presents a fertile ground for active acoustic sensing. However, despite significant advances, ensuring its robust and reliable deployment in the wild remains a key challenge. In this paper, we point out and articulate three critical issues that hinder the practicality of active acoustic sensing: its susceptibility to irregular or self-induced movements, vulnerability to interference from unrelated actions, and limited ability to perform fine-grained activity recognition across a diverse range of users. To address these challenges, we present several forward-looking perspectives, suggesting that overcoming these challenges requires not only advancements in algorithms but also the selection of tasks better suited to acoustic sensing, ultimately unlocking its full potential and ensuring stable performance. Our work aims to spark deeper discussions on reliable and robust active acoustic sensing solutions.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**; *Ubiquitous and mobile computing systems and tools*.

Keywords

Acoustic sensing, Wearable, In-the-wild

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1 Introduction

Sound is a fundamental medium for human perception, making microphones and speakers essential components in modern smart devices. This widespread integration has, in turn, spurred significant research interest in acoustic sensing technologies. The core idea is that these devices are configured to emit inaudible acoustic signals through the speakers. These signals are then reflected by targets in the environment and captured by the microphones on the device. By analyzing the received signals, we can extract information about the targets, such as their location and movement. The research community has explored the potential to extend the primary functionality of billions of devices from simple audio-based interactions to a wide range of intelligent acoustic sensing applications. These applications include, but are not limited to, activity recognition [1, 3, 4, 7, 15, 24], vital sign monitoring [16, 18], and environmental sensing [2, 17].

Despite rapid advancements in hardware, coupled with the growing prevalence of connected devices, provide a unique opportunity to enhance the accuracy of these systems. However, as we push acoustic sensing from controlled laboratory environments to real-world applications, we encounter significant practical challenges that threaten its widespread adoption. Therefore, in this paper, we highlight several prominent challenges that persist in current research, including (a) susceptibility to irregular user movements, (b) interference from unrelated actions, and (c) the difficulty of achieving fine-grained activity recognition across diverse user populations. At the same time, to address these limitations, we propose a series of forward-looking perspectives. We argue that overcoming these challenges requires not only advancements in algorithms, but also a thoughtful selection of tasks that align more closely with the inherent strengths of acoustic sensing. Only by doing so can we unlock the full potential of this technology and ensure its stable and reliable performance in the real world environment.

We summarize the contribution of this work as follows:

- We highlight several key challenges currently faced in active acoustic sensing.
- In response to the challenges outlined above, we offer a set of potential directions and perspectives to guide future research.
- We advocate for rethinking the capabilities of active acoustic sensing through the lens of task suitability.

2 Background

Acoustic sensing in smart devices [6, 8, 9] has emerged as a significant area of interest, driven by the ubiquitous presence of embedded speakers and microphones. This technology utilizes inaudible sound waves emitted by speakers, analyzing the resultant echoes captured by microphones to enhance human computer interactions (HCI). Compared to computer vision-based methods, active acoustic sensing offers significant advantages in energy efficiency while inherently preserving user privacy. When contrasted with RF signal-based approaches, it demonstrates greater generalizability and portability. Moreover, it does not require direct contact with the skin, making it naturally well-suited for continuous use in wearable devices.

Given these advantages, a variety of wearable applications have been developed based on active acoustic sensing, including gesture recognition [5, 21, 22, 24], activity tracking [8, 11, 20], and user authentication [10]. These applications leverage the acoustic characteristics of the environment along with the device's ability to detect subtle variations in sound wave patterns, thereby enabling fine-grained human-computer interaction tasks. Despite these exciting advances, we observe that the performance reported under certain conditions in existing studies remains less than satisfactory. Motivated by this, our work aims to explicitly identify these specific challenges and offer new perspectives for addressing them, focusing on both algorithmic improvements and task-alignment considerations.

3 Key Challenges

In this section, we present the challenges we have identified and analyze the reasons why they arise in practical active acoustic sensing scenarios.

3.1 User Movements (C1)

Deploying wearable active acoustic sensing systems in real-world environments inevitably involves various user-induced body movements. When such movements become excessive or unpredictable, they can significantly impair the performance of acoustic sensing, as evidenced by prior works. This is because, at its core, an active acoustic sensing system operates by transmitting a known signal $s(t)$, receiving its reflections from the surrounding environment as $r(t)$, and analyzing characteristics such as time delay, amplitude, and frequency shifts in $r(t)$ to infer information about the target (e.g., location or movement). Under ideal conditions:

$$r(t) = \sum_{i=1}^N a_i \cdot s(t - \tau_i) \quad (1)$$

When user movement occurs—whether it stems from intentional target actions or unintended, noise-inducing motions—it not only alters the distance between the system and the target, thereby affecting the time-of-flight delay, but also significantly impacts multipath propagation. Under such conditions, the received echo can be modeled as:

$$h(t, \tau) = \sum_{i=1}^{N(t)} a_i(t) \cdot \delta(\tau - \tau_i(t)) \quad (2)$$

Therefore, when the noise introduced by vigorous user movements significantly outweighs the signal generated by the target activity, it can overwhelm the desired features, making it difficult to extract meaningful information related to the intended actions. Although some researchers have applied specially designed signal processing techniques to suppress noise in mobile device-based sensing [14], these efforts have primarily focused on scenarios involving slow and predictable device movements.

3.2 Unrelated Actions (C2)

In addition to the interference caused by vigorous movements occurring simultaneously with target actions, numerous unrelated movements frequently occur during periods without any target activity—a common scenario given that many sensing tasks involve intermittent target actions. These irrelevant movements are diverse, challenging to comprehensively recognition, and can induce responses in the received acoustic signals, thereby complicating the sensing process. Consequently, when deploying acoustic sensing systems in the wild, a large number of irrelevant motions may lead to false recognition (e.g. if the intended task is to monitor users' facial expressions, actions such as drinking water may also produce noticeable spectral responses, resulting in these irrelevant behaviors being mistakenly identified as target activities).

Although recent works inspired by advances in the machine learning community have successfully reduced the false recognition rate below 20% through innovative algorithmic designs, these results are often limited to in the semi-wild [3, 4, 15]. Moreover, unlike computer vision, which benefits from massive available datasets, acoustic sensing often lacks such extensive data, potentially limiting the direct applicability of some data-driven methods. Furthermore, even a relatively small proportion (e.g., around 5%) of irrelevant actions being falsely recognized can substantially impair the user experience during real-world deployment, particularly in scenarios where irrelevant actions significantly outnumber target activities.

3.3 Variations Across Users (C3)

While active acoustic sensing exhibits robust cross-user performance in some coarse-grained activity recognition tasks [15], its ability to generalize across users is substantially limited in fine-grained wearable applications. The reason is that differences in body shape, movement styles—even when performing the same action—and skin properties all lead to highly individualized echo responses. This variability results in pronounced inconsistencies in the received signals across different users, complicating the process of learning user independent models. Unlike computer vision, which benefits from high-level semantic features with strong interpretability that facilitate generalization across individuals and tasks, features extracted from acoustic spectrograms are typically lower-level and less interpretable. This makes it more difficult to bridge the gap between diverse user characteristics. Although collecting a small amount of registration data from new users is currently acceptable for fine-grained HCI activity recognition tasks [4, 11, 24], user willingness to participate may decrease significantly as the number of classes increases or the perceived necessity of the task

diminishes. Therefore, this issue continues to pose a significant limitation to the application of acoustic sensing on wearable devices.

4 Potential Perspectives for Solutions

In this section, we aim to provide solutions for advancing acoustic sensing into real-world (in-the-wild) scenarios, drawing on multiple perspectives.

4.1 Advanced Machine Learning Methods

With ongoing advances in the field of machine learning, an increasing number of novel methods are providing valuable insights. By carefully selecting and applying appropriate machine learning techniques, it becomes possible to address some of the key challenges facing acoustic sensing.

- **Motion interference:** one promising approach is the use of multimodal fusion methods. While acoustic sensing excels at contactless motion perception, it can be susceptible to unstable sensing results due to various disturbances. Therefore, integrating it with more stable modalities—such as IMUs or passive acoustic sensing—can be an effective strategy. By combining information from these complementary sources, the impact of motion-induced interference can be significantly reduced (e.g. IMUs can provide information about device movement [13]). In addition, another promising direction is inspired by advancements in speech enhancement [20, 23]. In this domain, neural networks are trained to reconstruct the original spectrogram from inputs with artificially injected noise, thereby learning effective denoising capabilities that can be leveraged during inference. By treating motion-induced interference as "noise" and the target actions to be recognized as the "clean audio", this approach enables the training of encoder-decoder denoising networks that can effectively reduce the impact of such noise. In this way, motion artifacts can be suppressed, improving the robustness of activity recognition in acoustic sensing systems.
- **Unrelated actions:** one possible approach is to introduce a "Null" class. By collecting data on various irrelevant actions, the original target classification task can be expanded to include an additional category, enabling the system to distinguish between target and unrelated activities. Furthermore, given that it is impractical to collect data for all possible unrelated actions in real-world scenarios, out-of-distribution (OOD) detection methods can be employed to filter out previously unseen categories [12]. Specifically, these approaches identify and filter samples of irrelevant actions that fall outside the distribution of the training data, thereby improving the robustness of the system to unknown or unexpected activities.
- **Variations across users:** While transfer learning remains a commonly considered approach, unsupervised learning methods may also offer valuable support for this challenge. In addition, the integration of large language models (LLMs) [19] is a potential direction that warrants further exploration for interpreting low-dimensional spectrogram features, and may help models gain a deeper or more meaningful understanding of the underlying spectral semantics.

4.2 Task Suitability in Acoustic Sensing

Although advanced machine learning techniques may offer promising solutions for enabling acoustic sensing in the wild, these approaches still require ongoing exploration and refinement. Therefore, under current technological constraints, selecting tasks that are inherently well-suited to acoustic sensing remains crucial for realizing its full potential.

First, we should prioritize fine-grained sensing tasks that do not require continuous operation but still demand high levels of user accessibility. Typical examples include on-demand gesture control for smart home devices, speech enhancement during voice calls, and silent speech input for controlling smart glasses. These scenarios share three ideal characteristics:

- **User-initiated activation:** The sensing module can remain dormant most of the time and is actively controlled or explicitly triggered by the user.
- **High usability requirements:** The system should ensure low power consumption, unobtrusive operation, and strong privacy protection for the user.
- **Fine-grained recognition accuracy:** The system should be capable of fine-grained recognition, such as distinguishing finger-level gestures or subtle lip movements, often at a distance.

Active acoustic sensing is particularly well-suited for such tasks because (i) the on-demand activation mechanism ensures that the system is not exposed to excessive interference; (ii) microphones and speakers can be unobtrusively integrated into a wide range of devices; and (iii) its short wavelengths enable centimeter-level spatial resolution even at distances of one to two meters. By selecting appropriate tasks, we can design practical applications that exhibit stable and robust performance.

Second, for scenarios that require continuous operation, such as 24/7 health monitoring, occupancy detection, or hands-free AR/VR interaction. We should prioritize tasks that are already supported by reliable signals from other modalities, but still benefit from the complementary information that ultrasound uniquely offers. In practice, this means selecting applications where a low-power, always-on sensor (e.g., an IMU, barometer, passive acoustic sensor, or RFID tag) provides coarse-grained cues, while ultrasonic echoes offer the fine-grained, centimeter-level resolution that the primary modality cannot achieve. For example, for respiration behavior recognition, passive acoustic sensing can provide stable and fundamental information, while ultrasound can refine the signal by capturing subtle movements, further enhancing recognition performance. For hands-free AR/VR interaction, an IMU device can support hand-level gesture for basic control, while ultrasound enables precise finger-level gesture, improving the expressiveness and usability of the interface. By aligning ultrasound's strengths with complementary sensing modalities, we can achieve the stable, privacy-preserving, high-resolution performance demanded by always-on wearable applications.

5 Conclusion

In this paper, we have highlighted several critical challenges facing active acoustic sensing, particularly its susceptibility to user movements, interference from unrelated actions, and limited cross-user

performance. Despite significant advancements in hardware and algorithms, these issues continue to hinder the robust and reliable deployment of acoustic sensing systems in real-world environments. We have also proposed several forward-looking perspectives for addressing these challenges, emphasizing the importance of machine learning methods and task suitability. By selecting machine learning methods and tasks that align with the unique strengths of acoustic sensing, we can improve the system's robustness and accuracy. Ultimately, as we continue to refine the algorithms and select appropriate application scenarios, active acoustic sensing has the potential to unlock significant advancements in wearable sensing technology. We hope this work sparks deeper discussions and collaborative efforts aimed at pushing the boundaries of acoustic sensing in the wild.

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